10.2 The Stacked-Ball Effect

WORKSHEET

The question

STEP II 2010, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q10

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself From Section 10.1: if a mass M moving at speed u hits a stationary mass m elastically, what is the speed of m afterwards?

STAGE 1 The speed limit

Ask yourself How does $\frac{2b}{b+1}$ compare with 2? What happens as $b \to \infty$?

STAGE 2 Passing it along a chain

Ask yourself In part (ii), each particle is λ times as heavy as the next one. What does part (i) say about each collision in the chain?

Ask yourself Is $\frac{2\lambda}{\lambda+1}$ bigger or smaller than 1?

Try this next

1. **The real thing.** Try STEP I 2014, Question 10 (step.maths.org/questions/14-s1-q10): the stacked-ball problem with real numbers and imperfect bounces.
2. **An inelastic chain.** Try STEP I 2017, Question 10 (step.maths.org/questions/17-s1-q10). It sends a collision along a chain of particles whose masses grow geometrically. When is each particle involved in exactly two collisions?
3. **Three balls.** Three perfectly elastic balls are stacked, each much lighter than the one below. Show that the top ball leaves at $7U$ and rises to $49H$. With k balls, show that the top one leaves at $(2^k - 1)U$.
4. **Where does the energy come from?** In part (ii), show that the total kinetic energy is conserved even though A's speed is unbounded. What happens to the heavy particles?

Further reading

- **David Morin, *Introduction to Classical Mechanics*** (Cambridge University Press), Chapter 5, which includes the stacked-ball problem.
- **W. R. Mellen, "Superball rebound projectiles"**, *American Journal of Physics* 36 (1968). Short and fun.

- **Gary Flandro, "Fast reconnaissance missions to the outer solar system utilizing energy derived from the gravitational field of Jupiter"**, *Astronautica Acta* 12 (1966). The paper that proposed the Voyager "grand tour".

The full guided solution, and what the question leads to at university, is in Section 10.2:
https://beyondstep.keithmcnulty.org/ch10/stacked-balls.html